Section 9:

D. Flue gas analysis:

Boiler flue gas analysis is used to determine combustion efficiency.	
Carbon Dioxide (CO_2)	Indicates complete combustion
Carbon Monoxide (CO)	Indicates incomplete combustion
Oxygen (O_2)	Indicates the presence of excess air
Oxides of Nitrogen (NO_x)	A product of high temperature combustion
Combustibles	Material that burns when exposed to oxygen and heat

It is typical to target oxygen levels of 8% in low fire and 3% in high fire for gas fired burners. It is typical to target oxygen levels of 6% in low fire and 4% in high fire for oil fired burners.

Johnston Boiler Company recommends no greater level than 200 ppm of Carbon Monoxide in its burner operation. The acceptable “Industry Standard” level is 400 ppm or less.

Johnston Boiler Company recommends zero combustibles for a gas fired burner.

Johnston Boiler Company recommends a maximum #2 Smokespot (Ringelmann Chart) in its oil fired burner.

Air Properties:
For a burner originally adjusted to 15% air, changes in combustion air temperature and barometric pressure cause the following in excess air:

Air Temperature	Barometric Pressure (In. HG)	Resulting Excess Air %*
40	29	25.5
60	29	20.2
80	29	15.0
100	29	9.6
120	29	1.1
80	27	7.0
80	28	11.0
80	29	15.0
80	30	19.0
40	31	34.5
60	30	25.0
80	29	15.0
100	28	5.0
120	27	-5.5

* Expressed as a percent of the Stoichiometric air required.

Section 10: Maintenance

A. General

A good maintenance program is essential to the lasting operation of your Johnston burner. Properly maintaining the burner will reduce unnecessary downtime, lower repair costs, and increase safety.

All electronic and mechanical systems require periodic maintenance. Automatic features do not relieve the operator from maintenance duties.

Any unusual noise, improper gauge measurement, or leak can be a sign of a developing problem, and should be taken care of immediately.

B. Component-Specific Maintenance

Most of the components and systems of your Johnston burner require little maintenance other than regular inspection. Cleanliness is essential to the inspection of the burner. Keep the exterior free of dust. Keep the controls clean. Clean dust and dirt from the motor starters and relay contacts.

See individual component manuals for more information on specific component maintenance issues.

C. Maintenance Schedule

Johnston Boiler Company recommends the following maintenance schedule for the upkeep of your Johnston burner. Any omission form this schedule could cause the Warranty to be voided. Only qualified and experienced boiler, burner, and control personnel should attempt any maintenance and/or repair on any boiler/burner. Be sure you have read and understood this entire manual before attempting maintenance or repair.

This maintenance schedule assumes a Johnston burner. Refer to your boiler manual for trim and boiler maintenance.

Daily:
The following maintenance tasks should be performed daily by the operator:

1. Check boiler manual for all items pertaining.
2. Check over the boiler and burner for proper operating pressures and temperatures.
3. Check the startup and operation of the burner pilot and main flames. Check the appearance of the flames for proper and stable combustion. Standard electrode setting is 1/8” to 3/16” gap.
4. For oil-fired burners, check the operation of the fuel oil pump, heater (for heavy oil) and air compressor. Check the oil level in the compressor. Check the condition of the fuel oil strainer for cleanliness.
5. Check to make sure that an adequate air supply into the boiler room is being maintained. Check to make sure that the inlet screen on the blower assembly is clean and free from obstruction.
6. Check for fuel and air leaks. Any leaks in the piping, fittings, controls, should be attended to promptly.
7. Replace failed indicating lights and ensure that all alarms are operational.
8. Maintain a clean, orderly, and safe boiler room.

Section 10:

Weekly:
The following maintenance tasks should be performed weekly by the operator:

1. Check the burner linkage joints, arms, and rods for tightness. Check the linkage firing rate motor (MOD motor), shafts, bearings, and flow control fuel valves for proper and smooth operation.
2. If a heavy oil burner, clean the oil gun assembly. In cleaning the oil nozzle, disassemble and clean with solvent and compressed air.
3. Grease the burner blower shaft bearings sparingly (only while in operation). See the manufacturer's recommendations for lubricating the blower motor.
4. If installed, check the boiler outlet flue gas damper for proper and smooth operation.
5. Check the peepsite and sightports for cracks in the lens.

Monthly:
The following maintenance tasks should be performed monthly by the operator:

1. Check the burner pilot assembly for cleanliness and proper condition. Check the electrode for cracks in the porcelain and proper adjustment. Check the condition of the lead wire, and its connectors. Standard electrode setting is a 1/8" to 3/16" gap.
2. Check the flame scanner for cleanliness. Check the condition of its connecting cable.
3. If a light oil fired burner, check the oil gun assembly for cleanliness and condition. Clean as may be required. In cleaning the oil nozzle, disassemble and clean with solvent and compressed air.
4. Check the blower assembly condition, cleanliness and proper operation, including the motor, blower wheel, air damper, and linkages.
5. If a heavy oil fired burner, check the operation of the steam trap for the oil preheater.
6. If a gas fired burner, check the main gas valves for proper, smooth operation.
7. If a combination fuel burner, check the fuel selector shifter mechanism for tightness and proper operation.
8. Check operation of the surface or continuous blowoff valve.
9. Change oil in air compressor after 500 hours of operation.

Annually:
The following maintenance tasks should be performed annually by the operator:

1. Check the condition of the burner internals, including the swirlers, main gas ring, oil gun assembly and burner refractory.
2. Check gauges and thermometers for proper calibration.
3. Open, clean and inspect the burner fireside. Inspect the condition of the refractory, Hairline cracking in the refractory is normal. Loose or missing pieces in the refractory must be patched, repaired, or replaced.
4. Check the condition of the fireside gasketing. Check for hardness and/or cracking. Replace fireside gasketing as needed, but not less than every three years.

Section 10:

BLOWER BEARING

Lubrication Instructions

All SEALMASTER bearing units are prelubricated with grease chosen for its chemical and mechanical stability. SEALMASTER bearings are designed for lubrication with grease. Units furnished with a grease fitting should be periodically relubricated. The relubrication interval depends on bearing operating conditions: speed, temperature, and environment. (See Conditions in Table 1 below for typical relubrication schedules.)

Figure 1 shows the annular passage and port feature of the SEALMASTER relubricable units. This feature provides positive means for relubrication while allowing several degrees of misalignment. If the grease fitting supplied with the unit is replaced, make certain the same length of thread engagement is maintained.

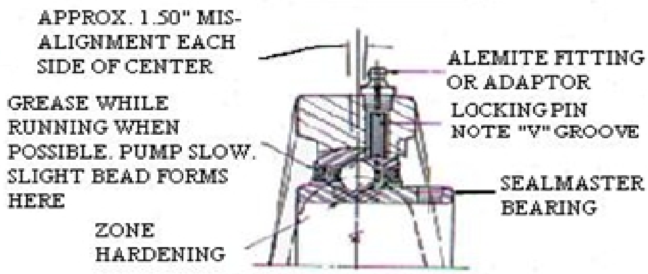

Fig. 1

Relubrication

1. Add grease slowly with shaft rotating until a slight bead forms at the seals

2. Relubrication is generally accompanied by a temporary rise in operating temperature until the bearing chamber is stabilized with the proper amount of grease.

3. If necessary to relubricate while bearing is stationary, refer to SEALMASTER catalog for maximum grease capacity for the size bearing.

4. For abnormal operating conditions of high temperature or abnormal environments, consult SEALMASTER Engineers.

TABLE 1

Speed	Temperature	Cleanliness	Greasing Interval
100 RPM	Up to 120 F	Clean	6 to 12 Months
500 RPM	Up to 150 F	Clean	2 to 6 Months
1000 RPM	Up to 210 F	Clean	2 Weeks to 2 Months
1500 RPM	Over 210 F	Clean	Weekly
Any Speed	up to 150 F	Dirty	1 Week to 1 Month
Any Speed	Over 150 F	Dirty	Daily to 2 Weeks
Any Speed	Any Temp.	Very Dirty	Daily to 2 Weeks
Any Speed	Any Temp.	Exterme Conditions	Daily to 2 Weeks

For normal operating conditions, relubricate with a grease conforming to NLGI No. 2 Penetration, free from chemical impurities, such as dust, rust, metal particles or abrasives.

Section 10:

JOHNSTON BOILER COMPANY

BOILER ROOM LOG

Date							
Time							
Boiler #							
Operator							
Water visible in gauge glass							
Combustion check (visual)							
Steam pressure							
Feedwater pressure							
Feedwater temperature							
Flue gas temperature							
Burner:							
Supply - Gas Pressure							
Valve - Gas Pressure							
Manifold - Gas Pressure							
Oil Regulated Pressure							
Oil Nozzle Pressure							
Oil temperature							
Atomizing media pressure							
Flame Signal							
Blowdown water columns							
Blowdown boiler							
Comments/Observations							

Section 11: Parts, Service and Warranty Claims

NOTE:
It is the responsibility of the individual, company, organization, or institution receiving a new burner, or packaged system to inspect the shipment for damage, alteration or missing components upon initial receipt. Transporters insist that all deficiencies be noted on the shipping documentation prior to acceptance. Johnston will not accept responsibility for shipping damages or claims against transporters.

A. Parts and Service Policy

Johnston Boiler Company markets their products through a distribution system, made up of independent manufacturers' representatives. For parts and service requirements, contact your authorized Johnston representative.

B. Parts Warranty Claims and Material Return

Johnston Boiler Company's warranty on parts, whether purchased as a component part of a new boiler or as a replacement part, is as described under the "WARRANTY AND EXCLUSION OF IMPLIED WARRANTIES" paragraph of Johnston's Standard Terms and Conditions of Sale. Johnston's parts warranty claim policy and return procedure is as follows:

It is recognized that often a warranty replacement part may be needed in advance of returning the claimed defective part for warranty consideration. Johnston Boiler Company can accommodate this situation. In order for Johnston to advance a warranty replacement part, a written purchase order must be issued to Johnston Boiler Company in the amount of the selling price for the replacement part and to cover any prepaid shipping charges that Johnston may incur. Upon receipt of the purchase order, and subject to availability of the replacement part, the advance warranty replacement part will then be shipped in the manner requested. At this same time, Johnston will issue and submit an invoice covering the part and freight charge.

Johnston will include a pre-numbered, 3-part Return Authorization Tag (Exhibit A) with any shipment of an advance warranty replacement part. This tag <u>must</u> be completed by someone knowledgeable of the claimed defect in the part to be returned. The green copy of the tag, 'TO BE RETAINED BY PARTY MAKING RETURN" is to be kept by the party making the return. The remaining two (2) copies of the tag yellow "JBC RECEIVING GIVE TO JBC SERVICE/PARTS" and buff (hard copy) "ATTACH TO PART BEING RETURNED (ALSO TO INCLUDE YELLOW COPY): are to be attached to the claimed defective part. The part is then to be returned to Johnston Boiler Company, freight prepaid (collect freight shipments will not be accepted), for warranty consideration.

Upon Johnston's receipt of the claimed defective part, the part will be reviewed for warranty consideration. A decision to accept or reject the warranty claim will be made. On parts that were manufactured by Johnston, this decision will be made by Johnston. On parts that were purchased by Johnston, the claimed defective part will be returned by Johnston to the actual manufacturer of the part, to be reviewed for warranty consideration. Any such other manufacturer will make the decision to accept or reject the warranty claim. Johnston will forward this decision onto its customer, the same as Johnston's own decision on warranty claims covering any parts it manufactures. Johnston will notify its customer of the decision on a parts warranty claim in writing.

On accepted parts warranty claims, Johnston Boiler Company will then cancel its invoice covering shipment of the advance warranty replacement part, less any prepaid shipping charges. Any prepaid shipping charges are due and payable by a customer, in accordance with the terms of Johnston Boiler Company's warranty.

On rejected parts warranty claims, Johnston Boiler will include in its written notification to a customer, the reason why the warranty claim was rejected. On rejected parts warranty claims, Johnston's invoice covering the warranty replacement part, whether advanced or otherwise is due and payable in the full amount by the customer.

All claimed defective parts must be returned to Johnston Boiler Company for warranty consideration no later than thirty (30) days after the advance warranty replacement part was shipped by Johnston.

On non-warranty claim parts returns, authorization to make any such return must be received from Johnston prior to making the actual return shipment. Johnston will mail to the customer the pre-numbered, 3-part Return Authorization Tag. This tag is then to be completed and handled in the same manner as a warranty claim.

Section 11:

Johnston Boiler Company **Return Authorization No__________________**

Return Freight Prepaid To:
Johnston Boiler Company
300 Pine Street
Ferrysburg, MI 49409

S.O. #____________________________________ Serial #_______________________

Model (or Catalog) #____

Job Name: ____

Location: ____

Date of Return: ______________________ By: ______________________________

This Item is being returned for the following consideration:

( ) Credit ( ) Warranty Replacement

( ) Credit – Warranty Replacement Already Received

Reason for Return: ____

____

____

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Exhibit A

Section 12: Troubleshooting

Johnston Burner with Johnston Boiler

PROBLEM	POSSIBLE CAUSE	SOLUTION
BLOWER DOES NOT START	Zero or Improper Power Supply Voltage	a. Check incoming power supply. Phase to phase voltage should match that which is shown on wiring diagram. b. Check disconnect switch if provided.
	Zero Voltage Between Flame Safeguard Power Supply Terminals. (Should be 115V).	a. Check control transformer primary and secondary fuses. Replace if blown. b. Make sure secondary voltage is near 115V. Replace transformer if voltage is wrong.
	Recycling Limit Circuit Open	a. Check voltage between recycling input and neutral. Voltage should be 115 V. b. Burner switch, Burner On relay, Gas On relay, or Oil On relay open. Applicable switch on contact must be closed. c. Operating Pressure or Temperature switch open d. Low water cut-off open e. f. Customer interlock not made. Any controls added to the recycling limit in the field must be made.
	Blower Motor Overload Relay Tripped	Check setting of overload relay against motor nameplate FLA. Correct any condition causing overload and reset overload relay.
	Blower Motor Fuses Blown	Replace fuses.
	Pre-Ignition Interlock Circuit Open	a. Check voltage between pre-ignition interlock input and neutral. Voltage should be 115. b. Make sure gas valve proof of closure switches are closed (if present). c. Make sure oil valve proof of closure switches are closed (if present).
	Blower Motor Contactor Defective	Replace Blower Motor Contactor.
	Blower Motor Defective	Replace Blower Motor.
	Flame Safeguard Defective	Replace Flame Safeguard.

Section 12:

PROBLEM	POSSIBLE CAUSE	SOLUTION
COMPRESSOR DOES NOT START (OIL BURNERS ONLY)	Fuel Selector Switch in Gas Position or Gas On Pushbutton is Depressed	Move switch to Oil position.
	Low Compressor Oil Levels Switch is Open	Close Low Compressor Oil Level Switch. (if supplied)
	Compressor Motor Overload Relay Tripped	Check setting of overload relay against nameplate FLA. Correct any condition causing overload and reset overload relay.
	Compressor Motor Fuses Blown	Replace Fuses.
	Compressor Motor Contactor Defective	Replace Compressor Motor Contactor.
	Compressor Motor Defective	Replace Compressor Motor.
OIL PUMP DOES NOT START (OIL BURNERS ONLY)	Oil Pump Switch not on (Heavy Fuel Only)	Turn Oil Pump Switch on
	Oil Pump Motor Overload Relay Tripped	Check setting of overload relay against motor nameplate FLA. Correct any condition causing overload and reset overload relay.
	Oil Pump Motor Fuses Blown	Replace fuses.
	Oil Pump Motor Contactor Defective	Replace Oil Pump Motor Contactor.
	Oil Pump Motor Defective	Replace Oil Pump Motor Contactor.
PILOT FLAME IS ESTABLISHED BUT FLAME SAFEGUARD FAILS TO SENSE IT	Pilot Flame is too Small	Increase Gas Pressure for Larger Flame
	Scanner Tube not Properly Aligned	Re-align Scanner Tube
	Cracked or Dirty Scanner Lens	Clean or Replace Scanner Lens
	Faulty Flame Scanner	Check Operation with a Candle or Lighter